

Hardy Wang

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TECHNICAL SKILLS

PCB Design (Altium): led UBC Solar steering wheel board and e-Radio RF sensor node from schematic through layout, fab, and bring-up
Bring-Up & Hardware Debug: traced an ADC and WiFi SoC resource conflict, and recovered a bad receiver clock with a planned 0 ohm rework option

Manufacturing Support: JLCPCB SMT builds with Basic-part selection, BOM review against design standards at Aecon

Component Sourcing: bench validated a drop-in buck converter replacement after the original went end of life

Embedded Firmware (C/C++, STM32, ESP32): wrote e-Radio board firmware, moved steering wheel debug from ST-Link to JTAG

Hand Assembly & Rework (THT/SMD): assembled and reworked steering wheel and e-Radio prototype boards by hand

Lab Measurement (Scope, DMM, Function Gen.): verified CAN/UART/I2C timing and power rails, characterized RF noise isolation

EDUCATION

University of British Columbia

Bachelor of Applied Science in Electrical Engineering

Vancouver, BC

Sept. 2024 – Dec. 2028

EXPERIENCE

Power and Signals Designer

Sept 2025 – Present

UBC Solar Car

Vancouver, BC

- One of ~8 engineers on the Power and Signals team; own the steering wheel board end-to-end and sync weekly with the mechanical and software subteams so the board's connector layout, mounting, and firmware interfaces land on schedule for vehicle integration.
- Simulated and laid out the board in Altium, then bench-validated it against system power and signal requirements with an oscilloscope, multimeter, and function generator, catching and fixing a noise-coupling issue before it reached the vehicle.
- Hand-assembled and reworked prototype and final boards (THT/SMD), driving the board from schematic to a reliable, race-ready unit.
- Evaluate and select components for the board, balancing performance, cost, and availability against team budget constraints.

Hardware Engineering Intern

May 2026 – Present

e-Radio Inc.

Toronto, ON

- Took a two-layer board from schematic through layout and routing, releasing it for fab and SMT assembly on schedule.
- Root-caused an ADC/WiFi radio resource conflict at the SoC level through hands-on bench debugging, then resolved it with a targeted firmware fix instead of a costly board respin.
- Designed and characterized the RF front end, isolating the receiver from high-noise digital nets at the layout level to keep signal integrity within spec.

Project Technician Intern

May 2026 – Present

Aecon Group Inc.

Toronto, ON

- Reviewed electrical Bills of Materials against engineering design standards, catching component mismatches before they reached procurement.

PROJECTS

Steering Wheel PCB | Altium Designer

Dec 2025 – Apr 2026

- Redesignated a steering wheel interface PCB implementing CAN, UART, and I2C communication and discrete driver inputs (horn, push-to-talk, regenerative braking, display controls).
- High-transient signals near the MCU risked corrupting driver input logic, so I isolated those lines with opto-isolators, dedicated return paths, and ground pour cutouts around the noise-sensitive nets, verified clean through bench testing.
- Migrated MCU programming and debug support from ST-Link to JTAG, verifying STM32 pin routing and signal integrity through bench testing.

Embedded RF Sensor Node | Altium Designer, ESP32, RF Front End

May 2026 – Aug 2026

- Designed a fleet-deployable sensor node (ESP32, Si4732 receiver, RTC, SD storage) from schematic through layout, JLCPCB fab and assembly, and bring-up, keeping the RF front end isolated from digital nets with ground pour cutouts.
- Planned 0 ohm DNP pads so the ESP32 could drive the receiver's clock if the crystal failed. When one unit's crystal backup failed, I populated them and validated data quality against reference boards, avoiding a respin.

Automotive Radar Perception Node | Altium Designer, STM32H7

Sept 2026 – Present

- Currently in schematic capture for a radar board around a BGT24MTR11 and an STM32H743VIT6, ferrite filtering the VDDA/VREF+ supply and scoping SPI, CAN-FD, and UART under a \$150 CAD parts budget.